

FRC Simulation Research

Inherited Baseline Audit and Framework Direction

CURRENT FOCUS

Characterize the inherited WarpX FRC baseline before building an independent, reproducible case-management workflow.

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This Week: Practical Takeover of the Inherited Baseline

MAIN PROGRESS

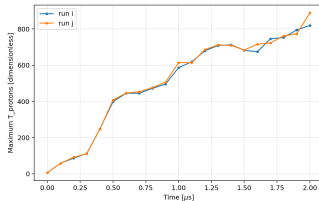
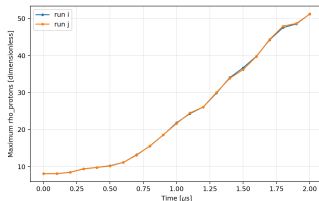
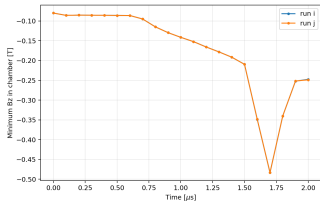
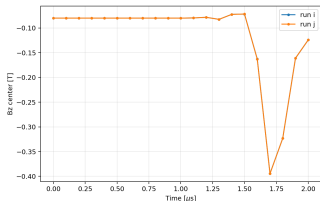
- Accessed the SCNet environment and identified the inherited FRC/WarpX working directories.
- Preserved the original directories as a read-only reference.
- Built a separate workspace: `FRC_jpsr`.
- Set up a Python analysis environment and compiled yt locally to read WarpX/AMReX plotfiles.

BASELINE STATUS

- Legacy executable: 3D WarpX, MPI + OpenMP, CPU backend.
- Two inherited reference runs: `i` and `j`.
- Both use the same physical input; differences are due to independent PIC realizations.
- Existing baseline duration: $2\ \mu\text{s}$, shorter than the imposed $4\ \mu\text{s}$ field-rise time.

Goal: turn the inherited case into a documented regression baseline before modifying or extending the workflow.

Scalar Audit: Runs i and j Agree Macroscopically



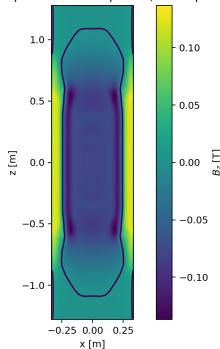
INTERPRETATION

- i and j are not bitwise identical, but their macroscopic evolution is almost the same.
- B_z and density are reliable first regression quantities.
- Pointwise T_i maxima are much noisier and should not be used as strict pass/fail criteria.
- Strong magnetic transient appears between 1.5 and 1.8 μ s.

Example run-to-run scale:
 $\max |\Delta B_{z,\text{center}}| \approx 3.3 \times 10^{-4} \text{ T}.$

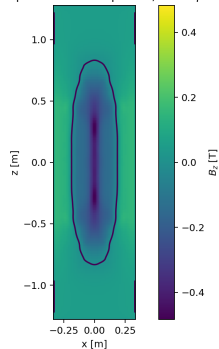
Spatial Audit: Development of a Reversed-Field Structure

B_z midplane slice -- step 1000, $t = 1.0 \mu\text{s}$



$t = 1.0 \mu\text{s}$: first closed $B_z = 0$ contour in the slice.

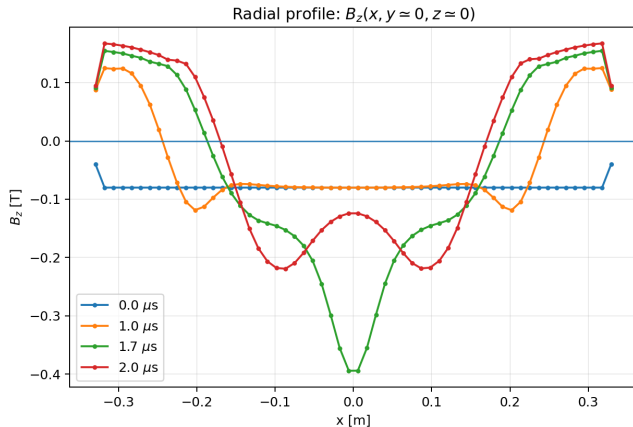
B_z midplane slice -- step 1700, $t = 1.7 \mu\text{s}$



$t = 1.7 \mu\text{s}$: strongest reversed-field compression in the available interval.

The contour shown is $B_z = 0$. It is a useful field-reversal proxy, but it is not yet a rigorous separatrix reconstruction. A proper separatrix estimate should later use the poloidal flux function $\psi(r, z)$.

Profiles Confirm the Reversal Is Spatially Extended



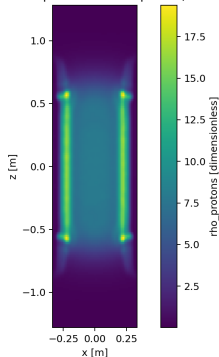
WHAT THE PROFILES SHOW

- Initially, B_z is negative and nearly uniform inside the plasma region.
- By 1.0 μs , radial and axial zero crossings appear.
- At 1.7 μs , the central negative field reaches its strongest value.
- At 2.0 μs , the reversed structure persists but partially relaxes.

The field reversal is a coherent spatial feature, not only a single-cell fluctuation or a scalar diagnostic artifact.

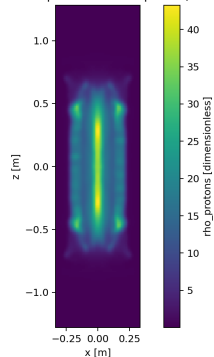
Density Compression Accompanies the Magnetic Evolution

rho_protons midplane slice -- step 1000, $t = 1.0 \mu\text{s}$



$t = 1.0 \mu\text{s}$: density begins to concentrate near the compressed region.

rho_protons midplane slice -- step 1700, $t = 1.7 \mu\text{s}$



$t = 1.7 \mu\text{s}$: strong density enhancement during magnetic compression.

In the central slice, ρ_{max} increases from about 8 initially to about 43 near $1.7 \mu\text{s}$. The full 3D maximum is larger, indicating that the strongest density point is not necessarily located exactly in the central $y \simeq 0$ plane.

Current Interpretation of the Inherited Run

OBSERVED SEQUENCE

bias field $\rightarrow$ field reversal $\rightarrow$ compression $\rightarrow$ partial relaxation

- No reversal at $t = 0$.
- Reversal proxy appears by $1.0 \mu\text{s}$.
- Strongest magnetic compression near $1.7 \mu\text{s}$.
- Structure remains visible at $2.0 \mu\text{s}$.

IMPORTANT LIMITATION

The existing run is a useful **regression baseline**, but it is not yet the final scientific formation case.

It ends at $2 \mu\text{s}$, while the applied field reaches its peak only at $4 \mu\text{s}$.

Next scientific target: a longer formation run that reaches and exceeds the full field-rise time, with topology diagnostics such as $\psi(r, z)$, R_s , L_s , and trapped flux.

Next Step: Framework v0.1

IMMEDIATE DEVELOPMENT TARGET

Build a minimal reproducible workflow that can generate, submit, and audit a controlled WarpX case without changing the validated physics.

First components:

- structured case configuration,
- input generation from a template,
- Slurm script generation,
- run manifest and provenance,
- smoke test with very few timesteps.

SHORT-TERM SEQUENCE

- ❶ Freeze the inherited baseline as read-only reference.
- ❷ Implement framework v0.1 around the same physics.
- ❸ Run a cheap smoke test.
- ❹ Reproduce the inherited 2000-step baseline.
- ❺ Then extend to a longer scientific formation case.

Summary: the inherited case is now a documented, physically characterized baseline rather than only a collection of old output files.

Summary

- The inherited WarpX/SCNet baseline has been located, preserved, and analyzed.
- Runs i and j show reproducible macroscopic FRC-like evolution.
- Scalar and spatial diagnostics indicate field reversal, compression, and partial relaxation within the first $2 \mu\text{s}$.
- The next phase is to build a small reproducible framework before launching new physics studies.

Thank you